

Customer-Constrained Optimization for Personalized Mortgage Rate Pricing

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Abstract—This paper presents a customer-constrained optimization framework for personalized mortgage rate pricing in retail banking. The objective is to recommend the lowest feasible mortgage rate for each application while preserving near-optimal expected profit. The framework integrates borrower-level features, market-relative acceptance modeling, and default risk estimation within a unified expected profit formulation.

Using a Freddie Mac-derived loan-level dataset spanning origination years 1999–2013, we demonstrate that the proposed approach achieves strong predictive performance ($AUC \approx 0.79$) and consistent optimization behavior across a heterogeneous borrower population. Empirical results show that the customer-constrained strategy reduces average mortgage rates by approximately 12.2 basis points, while incurring only a 3.92% reduction in expected profit.

These findings suggest that modest, data-driven adjustments to pricing can significantly improve customer outcomes while maintaining profitability. The framework provides a practical and interpretable approach for implementing customer-centric pricing strategies in competitive lending environments.

Index Terms—mortgage optimization, personalized pricing, credit risk, profit maximization, constrained optimization

I. INTRODUCTION

Mortgage pricing has traditionally relied on segmented approaches based on borrower characteristics. However, advances in machine learning have enabled more granular credit risk modeling and pricing optimization [1], [2].

This paper proposes a framework that determines the lowest feasible mortgage rate for each application while maintaining near-optimal expected profit. Unlike conventional segmentation-based approaches, the framework operates at the individual application level and explicitly accounts for market conditions through a market-relative acceptance model anchored to prevailing mortgage rates.

II. MORTGAGE APPLICATION INPUTS AND ASSUMPTIONS

Each mortgage application is represented by a feature vector:

$$x = (\text{credit score, income, DTI, LTV, loan amount, } \dots) \quad (1)$$

The model assumes:

- Credit score reflects borrower creditworthiness
- Income and debt-to-income (DTI) capture repayment capacity
- Loan-to-value (LTV) captures collateral risk
- Loan amount L directly impacts exposure

The cost of funds $c(x, r_m)$ is set as a spread below the prevailing market rate r_m , reflecting that funding costs are anchored to market conditions. Loss given default $LGD(x)$ is assumed to increase with LTV.

All optimization is performed at the individual application level.

III. PROBLEM FORMULATION

We formulate the mortgage pricing problem at the individual application level, where all quantities are defined with respect to a given borrower feature vector x , offered rate r , and prevailing market rate r_m . The feasible mortgage rate set is defined as a symmetric window around the market rate:

$$\mathcal{R}(r_m) = [r_m - 0.01, r_m + 0.02] \quad (2)$$

This ensures that the optimization remains anchored to contemporaneous market conditions, with rates spanning from 100 basis points below to 200 basis points above the prevailing market rate.

We define expected profit as the product of acceptance probability and conditional profit:

$$\Pi_{\text{exp}}(x, r, r_m) = P(A = 1 \mid x, r, r_m) \cdot \Pi(x, r, r_m) \quad (3)$$

where $P(A = 1 \mid x, r, r_m)$ denotes acceptance probability and $\Pi(x, r, r_m)$ denotes conditional profit.

A. Notation and Definitions

We summarize key variables used in the formulation:

- x : Feature vector representing a mortgage application
- r : Offered mortgage interest rate
- r_m : Prevailing market mortgage rate (PMMS)
- $\mathcal{R}(r_m)$: Feasible set of mortgage rates
- $P(A = 1 \mid x, r, r_m)$: Acceptance probability
- $P_d(x, r)$: Default probability
- $\Pi_{\text{exp}}(x, r, r_m)$: Expected profit
- $\Pi^*(x)$: Maximum achievable expected profit
- $r^*(x)$: Customer-constrained optimal rate
- ε : Profit tolerance parameter

IV. PROFIT MODEL

We define the profit from a mortgage loan as a function of the offered rate and borrower characteristics. The formulation captures interest income, expected credit losses, and acquisition costs.

$$\Pi(x, r, r_m) = L(r - c(r_m))T - P_d(x, r) LGD(x) L - C_{acq}(x) \quad (4)$$

where L denotes loan amount, T denotes loan term, $c(r_m) = r_m - 0.015$ represents the cost of funds as 150 basis points below the prevailing market rate, and $C_{acq}(x)$ represents borrower-specific acquisition or origination costs, including underwriting, processing, and distribution expenses.

A. Economic Interpretation of the Profit Function

The expected profit formulation captures three competing forces in mortgage pricing:

- **Interest Margin:** Higher mortgage rates increase per-loan revenue through the spread $(r - c(r_m))$.
- **Acceptance Behavior:** As the offered rate rises above market, borrower acceptance probability $P(A = 1 \mid x, r, r_m)$ declines, reducing loan origination volume.
- **Default Risk:** Higher rates may also increase borrower stress, leading to higher default probability $P_d(x, r)$.

These competing effects naturally induce a unimodal relationship between expected profit and mortgage rate, where profit increases initially with rate due to margin expansion, and then decreases due to declining acceptance and increasing default risk.

The trade-off between pricing and demand is well studied in revenue optimization literature [5].

V. BEHAVIORAL MODELS

We model borrower acceptance and default behavior as functions of both borrower characteristics and the offered mortgage rate.

The acceptance probability is defined as:

$$P(A = 1 \mid x, r, r_m) = \frac{1}{1 + \exp(-(\beta_0 + \beta^\top x + \gamma(r - r_m)))} \quad (5)$$

where r_m denotes the prevailing market mortgage rate and $(r - r_m)$ represents the offered-rate spread relative to market. A positive spread indicates the offered rate is above market, reducing acceptance probability when $\gamma < 0$. The parameter r_m is treated as exogenous and is set to the Freddie Mac Primary Mortgage Market Survey (PMMS) 30-year fixed rate corresponding to each loan's origination month [4].

This functional form ensures that probabilities remain bounded between 0 and 1 while capturing nonlinear sensitivity to the offered rate relative to market conditions.

The default probability is modeled as:

$$P_d(x, r) = \frac{1}{1 + \exp(-(\alpha_0 + \alpha^\top x + \eta r))} \quad (6)$$

A. Modeling Assumptions

The acceptance and default models are specified using logistic regression forms to capture nonlinear dependencies on borrower characteristics and mortgage rate. This specification captures the idea that borrower decisions depend not only on

the absolute offered rate, but also on how competitive the offer is relative to prevailing market conditions.

- The acceptance model $P(A = 1 \mid x, r, r_m)$ captures borrower sensitivity to pricing relative to market, where rates above the market rate reduce the likelihood of loan acceptance.
- The default model $P_d(x, r)$ captures credit risk, where borrower characteristics and pricing jointly influence repayment behavior.

While these models can be replaced by more sophisticated machine learning models (e.g., gradient boosting), the logistic specification provides interpretability and analytical tractability. The acceptance model is specified rather than directly estimated from data, as real-world datasets do not provide counterfactual pricing outcomes. The market-relative formulation partially mitigates this limitation by grounding the acceptance sensitivity in observable market conditions.

VI. CUSTOMER-CONSTRAINED OPTIMIZATION

We define the profit-maximizing rate as the rate that maximizes expected profit over the feasible set:

$$\Pi^*(x) = \max_{r \in \mathcal{R}(r_m)} \Pi_{\exp}(x, r, r_m) \quad (7)$$

$$r^*(x) = \min \{r \in \mathcal{R}(r_m) : \Pi_{\exp}(x, r, r_m) \geq (1 - \varepsilon)\Pi^*(x)\} \quad (8)$$

A. Interpretation of Customer-Constrained Optimization

The optimal rate $r^*(x)$ maximizes expected profit for a given application. However, this purely profit-driven strategy may lead to higher pricing that is not competitive or customer-friendly.

To address this, we introduce a constraint that allows for a small tolerance ε in expected profit. The constrained rate is defined as the lowest rate that achieves at least $(1 - \varepsilon)$ of the optimal profit.

This formulation reflects a practical business objective: reduce pricing where possible, while ensuring that the loss in profitability remains bounded.

From a decision-making perspective, this can be viewed as a form of *controlled revenue sacrifice* to improve customer outcomes, competitiveness, and potential long-term retention.

VII. SIMULATION STUDY

We evaluate the proposed framework using a Freddie Mac-derived mortgage dataset with heterogeneous borrower and loan characteristics.

A. Dataset and Empirical Setup

Experiments are conducted using a publicly available dataset derived from Freddie Mac loan-level data [3], containing borrower attributes, loan characteristics, and delinquency outcomes. The sample covers origination years 1999–2013, with prevailing market rates ranging from 3.57% to 8.52% and averaging 7.04%.

Default is defined as loans reaching 90 or more days past due ($\text{MonthsDelinquent} \geq 3$), consistent with industry standards.

Borrower features include credit score, debt-to-income ratio, loan-to-value ratio, loan amount, servicing characteristics, and property attributes. Additional feature engineering includes nonlinear transformations and interaction terms.

Since real-world datasets do not provide counterfactual pricing outcomes, borrower acceptance behavior is not directly observed. We therefore model acceptance as a function of the offered rate spread relative to the prevailing PMMS market rate, which is matched to each loan's origination month. The market rate r_m is sourced from the Federal Reserve Economic Data (FRED) repository [4]. While the acceptance sensitivity parameter is specified rather than estimated, anchoring the model to observed market rates provides an empirically grounded basis for the optimization.

B. Evaluation Metrics

We evaluate the framework using the following metrics:

- **Average Expected Profit:** Mean of $\Pi_{\text{exp}}(x, r, r_m)$ across all applications
- **Average Selected Rate:** Mean of recommended rates
- **Profit Retention:**

$$\frac{\Pi_{\text{exp}}(x, r^*(x))}{\Pi^*(x)}$$

- **Acceptance Rate:** Average $P(A = 1 \mid x, r, r_m)$

C. Optimization Procedure

For each application x :

- 1) Set the feasible rate grid $\mathcal{R}(r_m) = [r_m - 0.01, r_m + 0.02]$
- 2) Compute for each candidate rate $r \in \mathcal{R}(r_m)$:
 - $P(A = 1 \mid x, r, r_m)$
 - $P_d(x, r)$
 - $\Pi(x, r, r_m)$
 - $\Pi_{\text{exp}}(x, r, r_m)$
- 3) Compute optimal profit $\Pi^*(x)$
- 4) Select lowest rate $r^*(x)$ satisfying:

$$\Pi_{\text{exp}}(x, r, r_m) \geq (1 - \varepsilon)\Pi^*(x)$$

D. Evaluation

For each application:

- Evaluate a discretized grid of 141 candidate rates
- Compute expected profit at each candidate rate
- Compare pure-profit vs. constrained rate

VIII. RESULTS

We present empirical results demonstrating both predictive performance and optimization behavior of the proposed framework.

A. Predictive Model Performance

Gradient boosting methods such as XGBoost have demonstrated strong performance in credit risk modeling tasks [2]. A gradient boosting model (XGBoost) is used to estimate default probability based on borrower, loan, and servicing features. The model achieves an AUC of approximately 0.79 on held-out data, indicating strong predictive performance for loan-level risk estimation.

B. Profit vs Rate Behavior

Expected profit exhibits a unimodal relationship with respect to the mortgage rate, reflecting the trade-off between margin expansion and declining acceptance probability. The profit-maximizing rate sits above the market rate, where the interest margin is positive, while the constrained rate is shifted toward the market rate to improve customer outcomes.



Fig. 1. Expected profit as a function of mortgage rate for a sample applicant. The optimal rate sits above the prevailing market rate; the constrained rate is shifted left while maintaining near-optimal profit.

C. Aggregate Profit Behavior

To assess population-level behavior, we evaluate expected profit across 1,000 sampled applications.

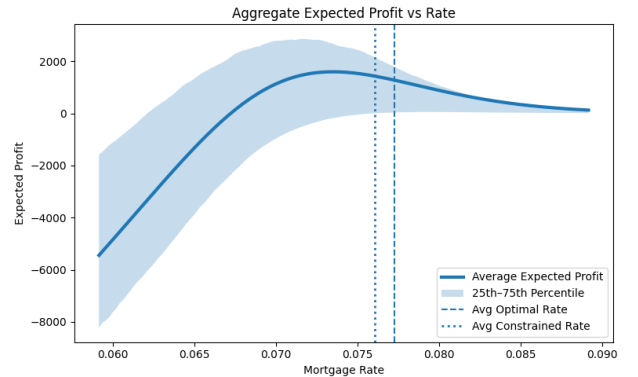


Fig. 2. Aggregate expected profit as a function of mortgage rate across 1,000 applications. The solid line represents mean expected profit, while the shaded region represents the interquartile range across borrowers.

The aggregate curve exhibits a clear unimodal structure, indicating a well-defined optimal pricing region. While individual applications display heterogeneity, the overall population behavior remains stable and predictable. These results indicate that traditional pricing strategies may systematically overprice lower-risk borrowers.

D. Impact of Customer-Constrained Optimization

As simple baselines, we consider (i) a fixed-rate pricing strategy where all applicants are offered the same mortgage rate, and (ii) a risk-tiered segmentation strategy where rates are assigned based on coarse credit score buckets. These approaches ignore finer borrower-level heterogeneity and do not account for prevailing market conditions. Compared to these baselines, the proposed framework enables individualized, market-anchored rate adjustments that improve both pricing efficiency and customer outcomes. While a full empirical comparison is beyond the scope of this study, these baselines highlight the limitations of non-personalized pricing approaches.

Across 1,000 sampled applications:

- Average market rate (r_m): 7.04%
- Average optimal rate: 7.73%
- Average constrained rate: 7.61%
- Average reduction: 12.19 basis points
- Average profit reduction: 3.92%

These results demonstrate that customer-friendly pricing can be achieved with only a 3.92% reduction in expected profit, while reducing rates by approximately 12.2 basis points. The optimal rate sits approximately 69 basis points above the market rate, reflecting a reasonable lender margin, while the constrained rate narrows this spread to approximately 57 basis points.

E. Rate Distribution Effects

The distribution of customer-constrained rates is consistently shifted toward lower values relative to profit-optimal rates, while maintaining substantial overlap.

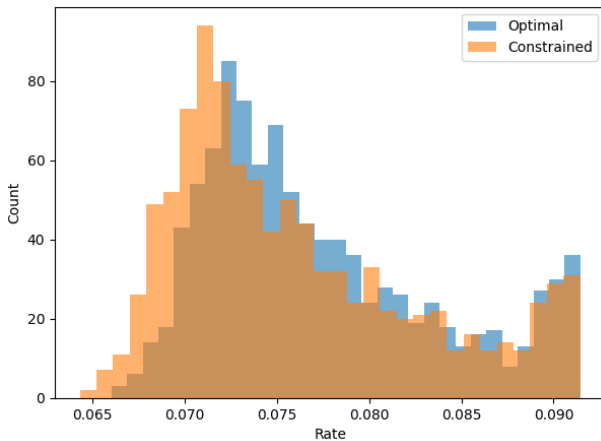


Fig. 3. Distribution of optimal vs customer-constrained rates across 1,000 sampled applications.

F. Robustness to Default Definitions

We also evaluated alternative definitions of default (e.g., ≥ 5 and ≥ 10 months delinquency) and observed similar qualitative behavior of the optimization framework across all threshold definitions.

G. Personalization Effects

Personalization primarily benefits lower-risk applicants, who receive reduced rates under constrained optimization, while higher-risk applicants exhibit minimal deviation from profit-optimal rates. This asymmetry arises because lower-risk borrowers tend to sit on flatter regions of the profit curve near the peak, allowing greater rate reduction before the ε -constraint binds. This preserves risk-adjusted profitability while improving customer outcomes for the borrowers most likely being overpriced under conventional segmentation strategies.

Incorporating market-relative pricing grounds the acceptance model in observable market conditions, ensuring that rate recommendations remain competitive across different interest rate environments spanning the 1999–2013 sample period.

IX. MANAGERIAL IMPLICATIONS

The proposed framework enables a shift from heuristic pricing to algorithmic decision-making at the individual application level. In practice, this allows lenders to systematically identify cases where pricing can be reduced without materially affecting profitability.

This is particularly relevant in competitive markets, where small pricing differences influence borrower choice. Rather than applying uniform discounts or broad segmentation rules, the framework provides a principled way to target rate reductions where they are most effective — namely, lower-risk borrowers whose profit curves exhibit sufficient flatness near the optimum.

From an operational perspective, the approach can be integrated into existing underwriting and pricing systems with minimal changes, as it relies on standard predictive models, a publicly available market rate series, and a simple grid-search optimization procedure.

X. CONCLUSION

This paper presents a customer-constrained mortgage rate optimization framework that balances profitability with customer competitiveness at the individual application level. By combining predictive modeling with constrained optimization and market-relative acceptance modeling, the framework identifies the lowest feasible rate that preserves near-optimal expected profit.

Using a Freddie Mac-derived loan-level dataset spanning 1999–2013, the proposed approach demonstrates strong predictive performance ($AUC \approx 0.79$) and consistent optimization behavior across a heterogeneous population of borrowers. Empirical results show that the customer-constrained strategy reduces average mortgage rates by approximately 12.2 basis

points while incurring only a 3.92% reduction in expected profit.

The results highlight the importance of incorporating both borrower behavior and market conditions into pricing decisions. The dependence of acceptance probability on the offered-rate spread relative to market introduces a natural competitive discipline that is not captured in traditional pricing frameworks. Traditional pricing strategies that ignore this effect may systematically overprice lower-risk borrowers, leaving measurable efficiency gains unrealized.

Several limitations remain. The acceptance sensitivity parameter γ is specified rather than directly estimated, as counterfactual pricing outcomes are not observable in the dataset. Future work may extend the framework by incorporating empirically estimated price elasticities from randomized pricing experiments, competitive market dynamics, richer borrower behavioral signals, and dynamic pricing strategies over the loan lifecycle.

Overall, the proposed approach provides a simple yet effective framework for customer-centric mortgage pricing and highlights the potential of combining machine learning with constrained optimization in financial decision-making.

XI. REPRODUCIBILITY AND CODE AVAILABILITY

To support reproducibility, the implementation of the proposed framework, including data preprocessing, model training, PMMS market rate integration, and optimization routines, is made publicly available.

The code repository includes:

- Data preprocessing and feature engineering pipelines
- PMMS market rate merge from FRED
- Default prediction model (XGBoost)
- Mortgage rate optimization framework
- Scripts for generating all figures in this paper

The code can be accessed at:

<https://github.com/srikanth421/mortgage-rate-optimization>

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